

EE340 Electromagnetic Theory

Assignment #05

Q. 1 Use MATLAB to plot the attenuation for the TE₁₀ mode of a waveguide with copper walls as a function of frequency. Do this for frequencies above the cutoff, keep in mind that R_s varies with frequency. Take $a = 2$ cm, $b = 1$ cm, $f_c = 10$ GHz, and assume that the waveguide is filled with a dielectric having $\epsilon_r = 2.25$.

Q. 2 A parallel-plate waveguide is formed by placing two infinite planar conductors at $y = 0, y = b$.

(a) Show that the electric field

$$E_x = E_0 \sin(\beta_y y) e^{-\gamma z}$$

defines a set of TE_{*n*} modes where

$$\gamma = \sqrt{\beta_y^2 - \beta_0^2} \quad \beta_0 = \omega \sqrt{\mu_0 \epsilon_0}$$

(b) For the modes of part (a), find the allowable eigenvalues, cutoff frequencies, and power transmitted, per unit width in the x direction.

Q. 3 Design a two-section binomial impedance transformer to match an empty ($\epsilon = 1$) X -band waveguide to a dielectric-filled ($\epsilon_r = 2.56$) X -band waveguide. Use two intermediate X -band waveguide sections, each quarter-wavelength long as shown in figure below. Assume dominant mode excitation, $f_0 = 10$ GHz, and waveguide dimensions of 2.286 cm $\times$ 1.016 cm.

[Note: You can self-study about the binomial impedance transformer for this numerical problem. The formulas for binomial transformer are as follows:

$$\Gamma_n = 2^{-N} \frac{\eta_L - \eta_0}{\eta_L + \eta_0} C_n^N$$

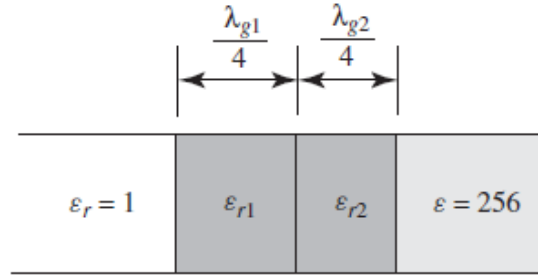
Where,

$$C_n^N = \frac{N!}{(N-n)!n!} \quad n = 0, 1, 2, \dots, N$$

(a) Determine the wave impedances of each section.

(b) Determine the dielectric constants of the lossless media that must be used to fill the intermediate waveguide sections.

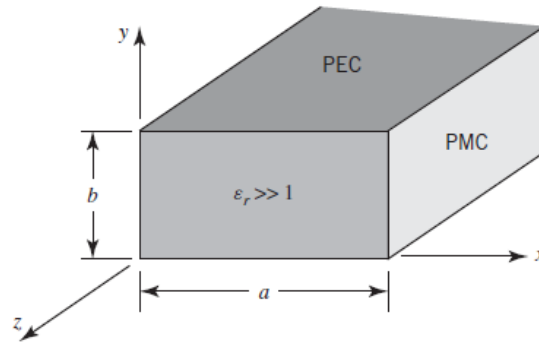
(c) Determine the length (in cm) of each intermediate quarter-wavelength waveguide section.



Q. 4 A rectangular dielectric waveguide with dimensions a and b ($a > b$), as shown in the figure, is used as a transmission line. The dielectric waveguide consists of a dielectric material with very high dielectric constant ($\epsilon_r \gg 1$). Also the waveguide has PEC (perfectly electric conducting) plates only on the top and bottom walls. The left and right walls are not covered with anything but can be treated as PMC (perfectly magnetic conducting) walls, where the tangential components of the magnetic field vanish.

Assuming TE^z modes only, determine:

- All of the allowable eigenvalues (β_x, β_y) for the TE^z modes for nontrivial solutions.
- The dominant TE^z mode and its cutoff frequency (in GHz) when $a = 0.9$ in. (2.286 cm), $b = 0.4$ in. (1.016 cm), $\epsilon_r = 81$ and $\mu_r = 1$.



Q. 5 The field between the plates is a linearly polarized (in the y direction) uniform plane wave travelling in the z - direction.

- Assume that the plates are perfect electric conductors. Then find the $\mathbf{E}$ and $\mathbf{H}$ field components between the plates. Neglect the edge effects of the finite plates.
- Find the separation d between the plates that creates resonance.
- Derive an expression for the Q of the cavity assuming a conductivity of σ for the plates. Neglect any radiation losses through the sides of the cavity.
- Compute the Q of the cavity when $f = 60$ GHz and $d = 5\lambda$ and 10λ . Assume a plate conductivity of $\sigma = 5.76 \times 10^7$ S/m.

